D 110235 **(Pages : 2)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS–UG) DEGREE EXAMINATION, NOVEMBER 2024

Physics/Applied Physics

PHY 5B 07/APH 5B 07—QUANTUM MECHANICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

Section A - Short Answer type

Answer **all** *questions in two or three sentences.*
Each correct answer carries a maximum of 2 marks.

1. What is meant by probability amplitude ?
2. Discuss uncertainty relationships for classical waves.
3. Define ultraviolet catastrophe.
4. Write down Einstein's photoelectric equation.
5. What do you mean by Bremsstralilung ?
6. State and explain Heisenberg uncertainty relationship.
7. What is meant by a free particle ? Write down the Schrodinger equation for a free particle.
8. Define zero point energy.
9. Explain the significance of Frank hertz experiment.
10. Discuss the fine structure splitting in Hydrogen atom.
11. Write down the Schrodinger equation cf the Hydrogen atom.
12. Discuss The limitations of Bohr atom model.

(Ceiling 20)

Section B – Paragraph / Problem type

Answer **all** *questions in a paragraph of about half a page to one page,*
each correct answer carries a maximum of 5 Marks.

13. X-rays with wavelength 1 A^o are scattered from a metal block. The scattered radiation is viewed at 90° to the incident radiation. Evaluate the Compton shift.
14. You are using a radiometer to observe the thermal radiation from an object that, is heated to maintain its temperature at 1278K. The radiometer records radiation in a wave length interval of 10.5nm. By changing the wavelength at which you are measuring, you set radiometer to record the most, intense radiation emission from the object. What is the intensity of the emitted, radiation in this interval ?

Turn over

15. Calculate the maximum wave length that hydrogen in its ground state can absorb. What would be the next maximum wavelength.

16. Estimate the minimum velocity that would be measured for a billiard ball (m = 80g) confined to a billiard table of dimension 1m.

17. Prove that i (d/dx) and d^2/dx^2 are Kemntian.

18. Calculate the expectation values of position and momentum of the particles trapped in the one dimensional box of width a.

19. Discuss the four quantum numbers of Hydrogen atom.

(Ceiling 30)

Section C – Essay type

Answer in about two pages, any **one** *question, correct answer carries 10 marks.*

20. Discuss the concept of wave packet. How is it represented analytically ? Also prove that the speed of the particle is equal to the group speed of the corresponding wave packet.

21. What is tunneling ? Write the theory of quantum mechanical tunneling. Write any three applications of tunneling.

(1 × 10 = 10 marks)